

Juniorprogrammierer.de

Java 1: Comments & Variables

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Agenda

- Syntax
- Comments I + II
- Variables I + II
- Homework

Syntax

- Other languages use {} to mark a block of code
- Indentations are used to make the code more readable
- Python uses indentations on the other hand to mark code blocks

Explanation >  syntax.py

```
1  if 5 > 2:
2      print("Five is greater than two!")
3  if 5 > 2:
4      print("Five is greater than two!")
5
6  if 5 > 2:
7      print("Five is greater than two!")
```

Comments I

- What are comments for
 - Explain code
 - Debug
 - Improve readability
- This is how you comment
 - 1 Line => `#This is a comment`
 - Multiple Lines => Python does not have it

```
#This is a comment  
print("Hello, World!")
```

```
print("Hello, World!") #This is a comment
```

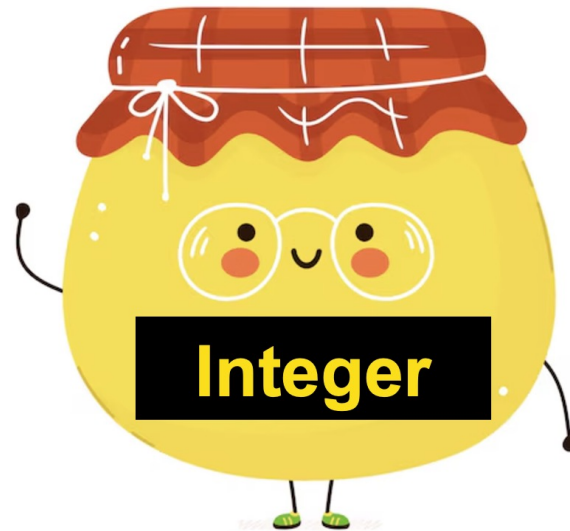
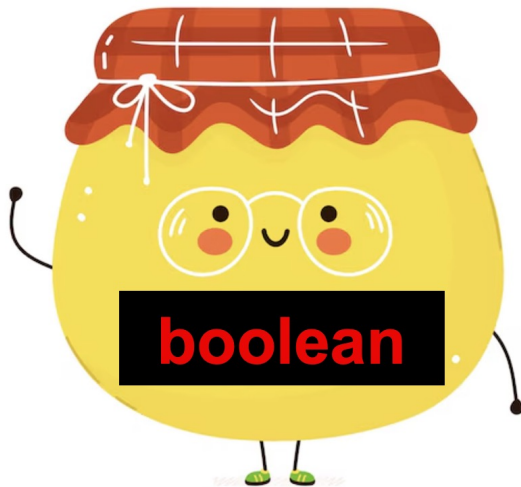
Comments II

- What do you think about this code example? Do the comments help here?
- Look forward to principles of clean coding ;)

```
15 # The 'print' function is a built-in function in Python
16 # that outputs text or other data to the standard output,
17 # which is typically the console or terminal.
18
19 print(
20     # This is the argument that is passed to the 'print' function.
21     # In this case, it is a string, denoted by either single ( ' ') or
22     # double quotes ( " "). Here, we've used double quotes.
23
24     "Hello, World!"
25
26     # The actual text within the quotes is the message that we want
27     # to display. The text "Hello, World!" is a traditional phrase
28     # often used by programmers as a simple test or demonstration project
29     # when learning a new programming language.
30
31     # This string is passed as an argument to the 'print' function.
32     # The 'print' function processes this argument and sends it to the standard output.
33
34     # Strings in Python can contain letters, symbols, numbers, and spaces.
35     # Here, we've used letters and symbols (comma, exclamation mark).
36
37     # By default, the 'print' function in Python adds a newline character
38     # at the end of its output, causing the next output to start on a new line.
39 )
```

Variables I

- Variables are a kind of data container for values or a jar ;)
- Variables = Type + Identifier + Value
- Assigning a type to a variable is called to declare



Variables II

- Python has no cmd to declare a variable, its assign indirectly
- A type of an variable can change simply by overwriting
- If you want to declare a certain type you can do it by casting
- `type(x)` method returns you the type of a variable
- Strings can be defined in single or double quotes
- Variable names are case-sensitive

```
41 x = 5
42 y = "John"
43 print(x)
44 print(y)
45
46 x = 4          # x is of type int
47 x = "Sally"   # x is now of type str
48 print(x)
49
50 x = str(3)     # x will be '3'
51 y = int(3)     # y will be 3
52 z = float(3)  # z will be 3.0
53
54 x = 5
55 y = "John"
56 print(type(x))
57 print(type(y))
58
59 x = "John"
60 # is the same as
61 x = 'John'
62
63 a = 4
64 A = "Sally"
65 #A will not overwrite a
```

Homework

- Create file SyntaxCommentVariable.py. Create 5 variables e.g. studentName ... (see rest in the pic) and create the following outcome

```
Student Name: John Doe  
Student ID: 15  
Student Age: 23  
Student Fee: 75.25  
Student Grade: B
```

- After you finished comment you solution out. Now change your file in a way that you get this outcome (Rectangle area is length * width and only use int types):

```
Length is: 4  
Width is : 6  
Rectangle area is: 24
```